Addis DemandFlow — Comprehensive Model Report

Addis DemandFlow · team_demandflow · Developer: Zedingil Zelalem · clock Africa/Addis_Ababa (EAT, UTC+3) · validation 18–31 Oct 2025

1. Headline

- Best model: **Ensemble: CatBoost + MLP (blended)**
- Models compared: **14** on the leaderboard + extended grid of **79** cells
- Features used: **57** leakage-safe features (events: 17, history: 15, calendar: 11, weather: 11, trend: 2, zone: 1)
- Rolling-origin folds: **4** (expanding, 14-day windows)
- Validation: raw RMSE **8.352**, MAE **5.738**, R^2 **0.904**; trimmed RMSE **8.352**, R^2 **0.904**

2. Feature engineering

57 features. Lags: lag_336h (14-day same-slot lag), lag_672h (28-day same-slot lag). Rain rolling sums: rain_3h_sum, rain_6h_sum, rain_24h_sum. Trailing zone windows: hist_zone_recent_7d, hist_zone_recent_14d, hist_zone_recent_28d, hist_zone_recent_56d.

feature	category	formula	rationale	known_at_forecast_time
zone_id	zone	dim_zone.zone_id	zone identity 1-12	yes
hour	calendar	pickup_datetime.dt.hour	diurnal demand cycle	yes
dayofweek	calendar	pickup_datetime.dt.dayofweek	weekday pattern	yes
day	calendar	day of month	pay-period effects	yes
month	calendar	month	seasonal trend	yes
weekofyear	calendar	iso week	within-year seasonality	yes
is_weekend	calendar	dayofweek>=5	weekend leisure demand	yes
is_payday	calendar	day in 1-3,28-31	salary spending pulse	yes
hour_sin	calendar	sin(2*pi*hour/24)	smooth time-of-day	yes
hour_cos	calendar	cos(2*pi*hour/24)	smooth time-of-day	yes
dow_sin	calendar	sin(2*pi*dow/7)	smooth day-of-week	yes
dow_cos	calendar	cos(2*pi*dow/7)	smooth day-of-week	yes
days_since_start	trend	date - 2025-01-01	captures growth trend	yes
temp_c	weather	weather temp (F fixed)	heat/comfort effect	yes
rain_mm	weather	previous-hour rain	rain demand effect	yes
is_raining	weather	rain_mm>0.05	rain yes/no	yes
heavy_rain	weather	rain_mm>7.6	heavy-rain flag	yes
rain_class	weather	cut(rain_mm)	non-linear rain response	yes
rain_3h_sum	weather	rolling 3h rain	sustained-rain effect	yes
rain_6h_sum	weather	rolling 6h rain	wet-spell effect	yes
rain_24h_sum	weather	rolling 24h rain	all-day wetness	yes
rain_is_raining_x_hour	weather	is_raining*hour	rain x time-of-day	yes
humidity_pct	weather	weather humidity	comfort	yes
wind_kmh	weather	weather wind	minor comfort	yes
has_event	events	interval join	event demand uplift	yes
event_count	events	overlap count	crowd of events	yes

feature	category	formula	rationale	known_at_forecast_time
has_sports	events	event_type	stadium match uplift	yes
has_concert	events	event_type	concert uplift	yes
has_conference	events	event_type	conference demand	yes
has_road_closure	events	event_type	detours/supply	yes
has_public_holiday	events	event_type	holiday demand shift	yes
has_school_break	events	event_type	school-holiday pattern	yes
has_exhibition	events	event_type	exhibition demand	yes
event_attendance	events	parsed crowd size	event size	yes
event_before	events	2h before	pre-event inflow	yes
event_during	events	during	in-event demand	yes
event_after	events	2h after	post-event dispersal	yes
has_sports_evening	events	has_sports * (hour>=17)	evening match uplift	yes
event_after_evening	events	event_after * (hour>=17)	evening dispersal peak	yes
hours_to_event	events	hours to next start	anticipation	yes
hours_since_event	events	hours since last end	dispersal decay	yes
lag_336h	history	trips 14 days ago	same slot last fortnight	yes
lag_672h	history	trips 28 days ago	same slot last month	yes
hist_zone_dow_hour_mean	history	expanding mean by zone/dow/hour	seasonal baseline	yes
hist_zone_hour_mean	history	expanding mean by zone/hour	diurnal baseline	yes
hist_zone_dow_mean	history	expanding mean by zone/dow	weekly baseline	yes
hist_zone_mean	history	expanding mean by zone	zone level	yes
hist_zdh_recent8w	history	8-week zone/dow/hour mean before cutoff	current regime profile	yes
hist_zh_recent8w	history	8-week zone/hour mean before cutoff	current regime diurnal	yes
hist_zone_recent_7d	history	mean last 7 days of history	recent level	yes
hist_zone_recent_14d	history	mean last 14 days of history	recent level	yes
hist_zone_recent_28d	history	mean last 28 days of history	recent level	yes
hist_zone_recent_56d	history	mean last 56 days of history	recent level	yes
rel_recent_7d	history	recent_7d / zone/dow/hour mean	recent-vs-profile level	yes
rel_recent_14d	history	recent_14d / zone/dow/hour mean	recent-vs-profile level	yes
rel_recent_28d	history	recent_28d / zone/dow/hour mean	recent-vs-profile level	yes
trend_index	trend	trailing city level, capped extrapolation	extrapolates growth	yes
active_drivers	operational	trip log	NOT available at forecast time	NO
avg_wait_min	operational	trip log	consequence of demand	NO
avg_fare_birr	operational	trip log	surge consequence	NO

3. Baselines (D1)

Model	train_time_s	RMSE	MAE	R2	MAPE_%	robust_RMSE	robust_R2
Seasonal-naive (zone/dow/hour mean)	0.0	11.2985	7.36	0.8252	26.43	11.2985	0.8252
Mean predictor (baseline)	0.0	27.5381	20.6891	-0.0381	163.89	27.5381	-0.0381

4. Model comparison (D2)

Model	train_time_s	RMSE	MAE	R2	MAPE_%	robust_RMSE	robust_R2
Ensemble: CatBoost + MLP (blended)	35.07	8.3523	5.7383	0.9045	28.82	8.3523	0.9045
CatBoostRegressor (tuned)	24.98	8.5079	5.7789	0.9009	26.8	8.5079	0.9009
CatBoost	34.67	8.5218	5.7861	0.9006	26.78	8.5218	0.9006
MLP	35.07	8.5849	5.9726	0.8991	31.9	8.5849	0.8991
ExtraTrees	21.44	8.699	5.8518	0.8964	26.13	8.699	0.8964
RandomForest	45.11	8.7285	5.8905	0.8957	26.42	8.7285	0.8957
HistGradientBoosting	21.38	8.7581	5.9391	0.895	26.95	8.7581	0.895
GradientBoosting	161.91	8.7749	5.9185	0.8946	27.01	8.7749	0.8946
XGBoost	35.39	8.8274	5.9624	0.8933	27.15	8.8274	0.8933
LightGBM	16.15	8.837	5.9854	0.8931	26.94	8.837	0.8931
Ridge	0.87	9.2087	6.4142	0.8839	38.77	9.2087	0.8839
Lasso	3.44	9.3731	6.4391	0.8797	37.68	9.3731	0.8797
KNN	3.21	9.7012	6.407	0.8712	28.76	9.7012	0.8712
DecisionTree	3.58	10.193	6.8528	0.8578	29.67	10.193	0.8578
Seasonal-naive (zone/dow/hour mean)	0.0	11.2985	7.36	0.8252	26.43	11.2985	0.8252
Mean predictor (baseline)	0.0	27.5381	20.6891	-0.0381	163.89	27.5381	-0.0381

5. Rolling-origin validation (D3) — 4 folds

Model RMSE 8.926 ± 0.660 vs seasonal-naive 11.782.

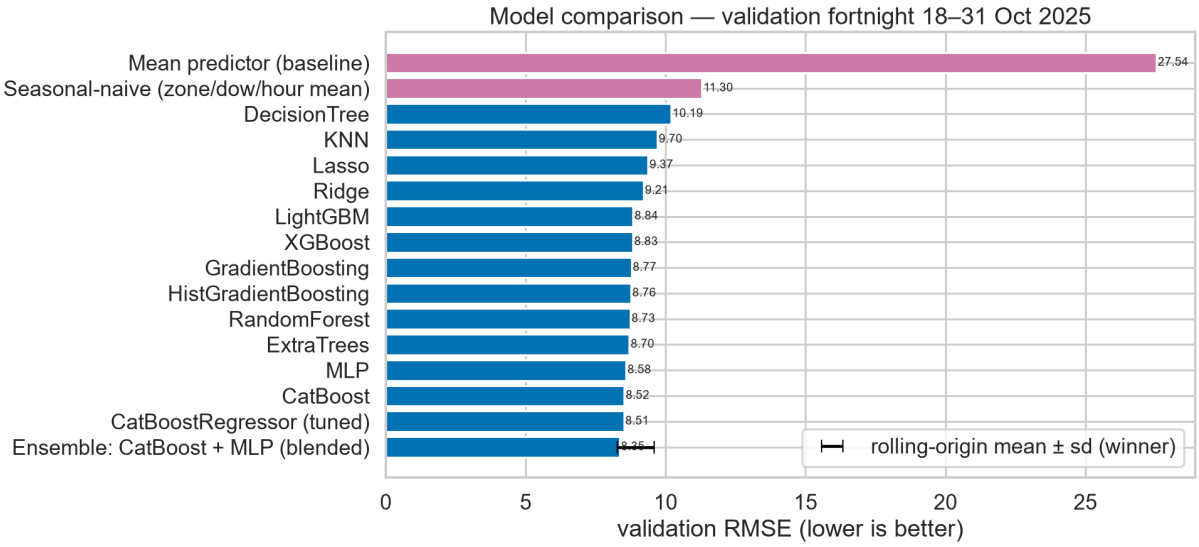
fold	val_window	model_RMSE	model_MAE	model_R2	model_robust_RMSE	model_robust_R2	naive_RMSE
2025-07-01	2025-07-01..2025-07-14	8.8696	6.0285	0.8718	8.8696	0.8718	10.727
2025-08-05	2025-08-05..2025-08-18	8.6809	5.7358	0.8864	8.6809	0.8864	11.4292
2025-09-10	2025-09-10..2025-09-23	9.8507	6.1848	0.8693	9.8507	0.8693	13.8674
2025-10-17	2025-10-17..2025-10-30	8.3038	5.7013	0.9043	8.3038	0.9043	11.105

6. Ablation (D5)

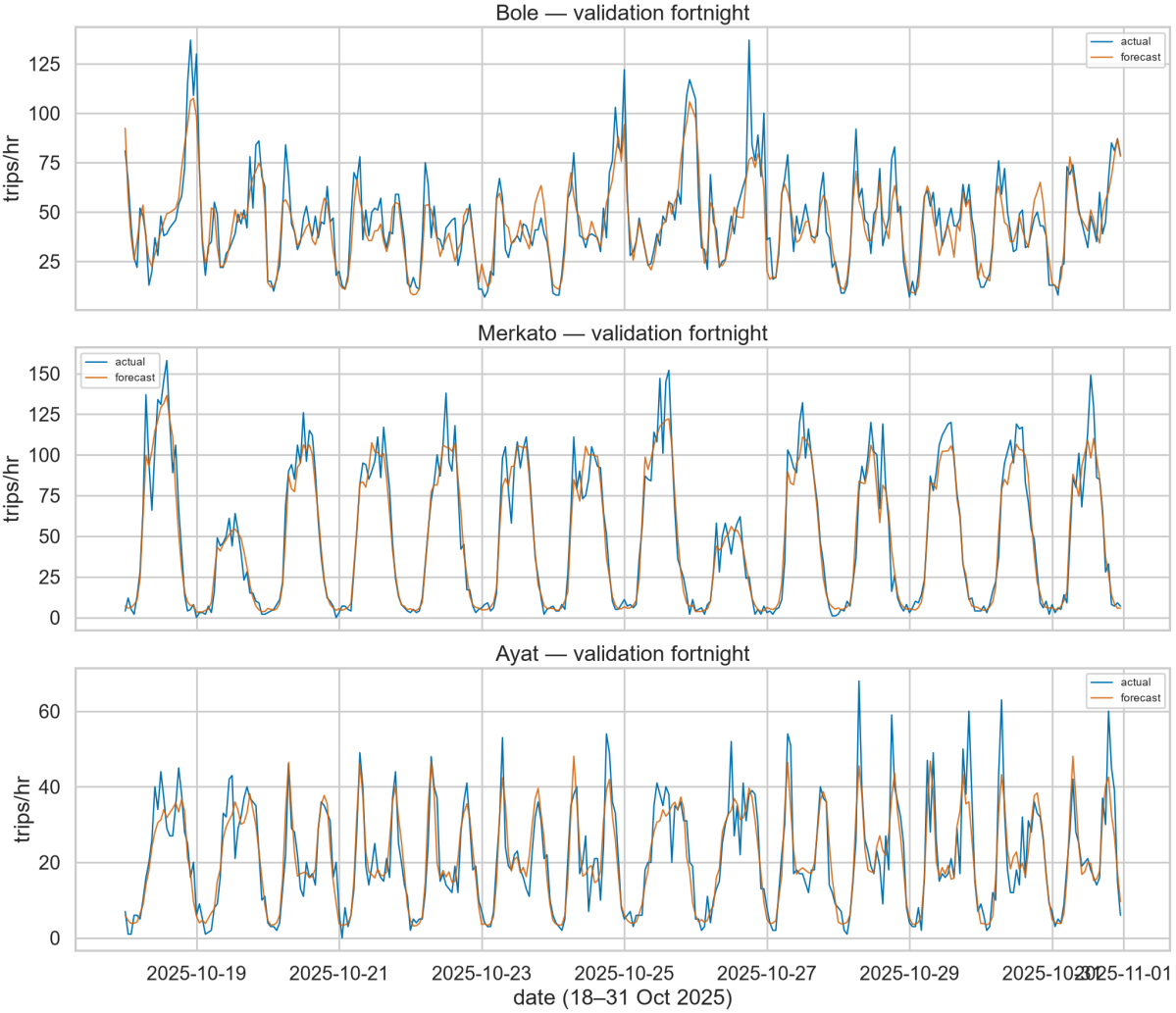
configuration	RMSE	MAE	R2	MAPE_%	robust_RMSE	robust_R2
calendar+zone+trend	9.663	6.3641	0.8722	28.41	9.663	0.8722
calendar+zone+trend+weather	9.2829	6.1732	0.882	28.56	9.2829	0.882
calendar+zone+trend+events	8.7197	5.9062	0.8959	27.2	8.7197	0.8959
calendar+zone+trend+weather+events	8.3858	5.7391	0.9037	27.86	8.3858	0.9037

8. Figures

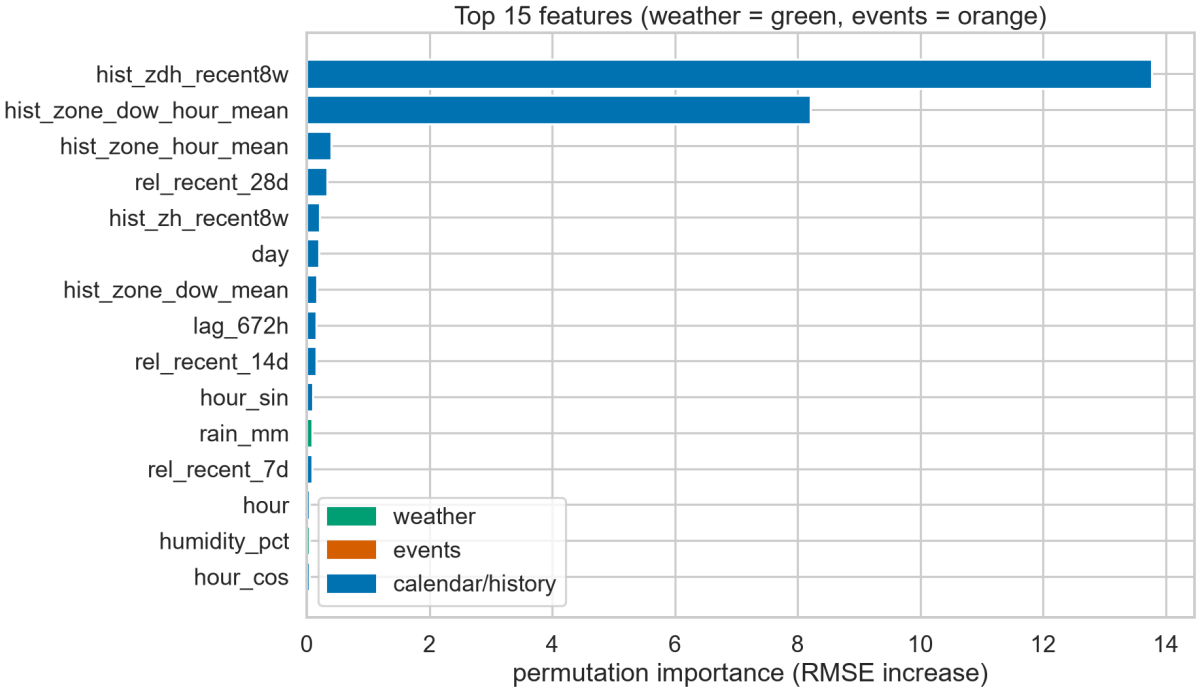
Validation RMSE of every model vs the two baselines.



Forecast vs actual hourly trips, three zones, full validation fortnight.



Top-15 permutation importances (weather=green, events=orange).


Top 15 features (weather = green, events = orange)
hist_zdh_recent8w
hist_zone_dow_hour_mean
hist_zone_hour_mean
rel_recent_28d
hist_zh_recent8w
day
hist_zone_dow_mean
lag_672h
rel_recent_14d
hour_sin
rain_mm
rel_recent_7d
hour
humidity_pct
hour_cos
weather
events
calendar/history
0
2
4
6
8
10
12
14
permutation importance (RMSE increase)